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Coordination Political Economy (CPE)

A Framework for K-Stability Economics

Treating coordination capacity as the primary scarce resource in complex societies

Definition

Coordination Political Economy (CPE) is a macro-institutional framework that treats coordination capacity—rather than output or consumption alone—as the primary scarce resource in complex societies. It combines game theory, network dynamics, and the K-Index to model how economies evolve relative to a trust threshold and how they fail through coordination collapse.

Where classical and neoclassical economics foreground prices, preferences, and equilibrium, CPE foregrounds:

- **$H_3(t)$** – coordination trust as a dynamic state variable
- **$K(t)$** – overall coordination capacity (K-Index)
- **θ** – critical trust threshold below which cascades and collapse become likely
- **$\lambda, R, \mu, \dots$** – modernization pressure, redundancy, learning, and other structural parameters

Economies are understood as ensembles of games under institutions, embedded in networks, subject to shocks and phase transitions.

Core Premises

1. Trust is a Factor of Production

Coordination trust (H_3) and institutional coherence (K) are treated as real, quantifiable inputs into production and governance, alongside labor, capital, and technology.

Implication: High output with collapsing H_3 is viewed as unsustainable extraction from the trust commons.

2. There is a Critical Threshold

Historical evidence and theory imply a critical trust density θ (≈ 0.37 – 0.38) at which a society's coordination network percolates or fails.

- **Above θ :** Positive feedback stabilizes cooperation
- **Below θ :** Cascades accelerate collapse

3. Economies are Dynamical Systems, Not Static Equilibria

Instead of solving for static equilibria, CPE focuses on trajectories:

$$dH_3/dt = f(H_3; \theta, \lambda, R, \dots)$$

$$dK/dt = g(K; H_3, \text{institutions}, \dots)$$

Central question: “Is the system moving toward or away from collapse?” not “Is the system at equilibrium?”

4. Game Forms are Policy Levers

Policies are evaluated by the game forms they induce: - One-shot vs repeated - Local vs global - High vs low transparency

And how those forms shape H_3 and K over time—not only by short-term GDP effects.

5. Collapse and Recovery are Part of the Model

CPE explicitly models: - Collapse velocity - Recovery probability - Intervention ROI

Using coordination laws as constraints on plausible trajectories.

Contrast with Classical Economics

Aspect	Classical/Neoclassical	Coordination Political Economy
Primary objective	Output, efficiency, utility	Coordination capacity, survival, coherence
Core variables	Prices, quantities, preferences	H_3 , K , θ , λ , redundancy, network structure
Stability notion	Static equilibrium	Dynamical stability vs threshold crossing
View of trust	Background assumption / externality	Explicit state variable & constraint
Failure modes	Market failure, recessions	Coordination collapse, phase transitions

CPE is designed to be **complementary**, not replacement: standard tools still matter, but they are nested inside a coordination-first lens.

Formal Core (Skeleton)

At its center, CPE uses a small set of “coordination laws”:

Threshold Law

There exists θ such that sustained $H_3 < \theta$ strongly predicts systemic collapse within a finite horizon.

Cascade Law

Below θ , the trust deficit $\Delta = (\theta - H_3)$ drives accelerating decline:

$$dH_3/dt \approx -\lambda_1 \Delta - \lambda_2 \Delta^2$$

Feedback & Percolation Laws

Above θ , trust and coordination reinforce each other; θ aligns with a percolation threshold of the coordination network.

Recovery & Intervention Regularities

Recovery probabilities and intervention ROI are asymmetric: prevention above θ has high returns; rescue far below θ is low-ROI and rarely succeeds.

These laws **constrain acceptable policy paths**, technological deployments, and institutional designs: trajectories that drive H_3 toward θ without compensating redundancy or trust-building are treated as dynamically unsafe, even if they raise short-term GDP.

Policy and Technology Implications

In CPE, economic and tech decisions are evaluated on three axes:

1. Effect on $H_3(t)$

Does this policy/technology strengthen or erode coordination trust (including Dark Trust components) over 5–50 years?

2. Effect on λ (modernization pressure) and R (redundancy)

- Does it accelerate cascades (higher λ)?
- Does it buffer them (higher R)?

Example: Hyper-centralized platforms may raise efficiency but lower redundancy.

3. Distance to Threshold / K-Stability

Does it push the system closer to θ , or deepen the safety margin?

“K-stable” designs maintain H_3 comfortably above θ under plausible stress scenarios.

Design Principles

Prefer Repeat-Play Structures

Repeated games with memory, stronger reputation systems, long-term contracts encouraged over one-shot, anonymous transactions.

Penalize Trust Predation

Actors who profit by eroding H_3 (disinformation, financial scams, extractive platforms) are treated like polluters of the trust commons.

Evaluate Tech on Full Impact

Large-scale AI and infrastructure projects assessed not just on efficiency gains but on their ΔH_3 , $\Delta \lambda$, and ΔR across time.

K-Stability Economics

Within Coordination Political Economy, **K-Stability Economics** is the specific formal apparatus that:

1. Uses the K-Index as a summary of coordination capacity
2. Applies the Coordination Laws to model collapse and recovery
3. Defines “K-stable” trajectories as those that keep H_3 and K safely above critical thresholds under stress

Put simply: - **CPE** = the broad field: “Economics grounded in the physics of coordination” - **K-Stability** = the concrete toolset: thresholds, laws, indices, and prediction machinery

The K-Index as Thermostat, Not God

What K should be used for:

Thermostat / Vital Signs - “Are we above θ or drifting toward it?” - “Are we burning trust faster than we can regenerate it?” - “Are interventions happening in the golden window or after the cliff?”

Constraint Role - “Don’t adopt policies that obviously push H_3 toward θ for short-term GDP” - “Don’t let λ explode without matching redundancy and trust”

Scenario-Testing - What game form does this policy create? - Are we pushing into PDs, races to the bottom? - Or into repeated, reputation-rich coordination games?

What K should NOT be:

Single Optimization Target - Goodhart’s Law: “When a measure becomes a target, it ceases to be a good measure” - Risk of “fake harmony,” censorship, data massaging

Technocratic Veto - Society contains plural values, different risk tolerances - K shouldn’t override legitimate value disagreements

Politically Captured Lever - Whoever controls K definition controls “what K says” - Measurement uncertainty creates abuse potential

Anti-Goodhart Safeguards

1. K as Compass and Guardrails, Not Dictator

Use K to diagnose and forecast, not as direct objective for agents.

2. Metric Ecology, Not Monoculture

Maintain plurality of metrics: - K-Index - Alternative civic indices - Random-audit qualitative panels

Never show a single magic number—use dashboards, comparisons, error bars.

3. Transparency and Contestability

- All schemas, weights, and code subject to open review
- Local autonomy and experimentation
- Multiple communities maintain their own views

4. Governance Guardrails

Explicitly encode: - “K is advisory, not infallible truth” - “No single metric may be sole basis for major coercive decisions” - “All metrics subject to challenge and forkability”

Dystopia to Avoid: K-Totalitarianism

A central planner or AGI using K as reward function could: - Decide dissent threatens K - Start “cleaning up noise” = authoritarian harmonization - Push apparent K up with coercion

This creates fragile Dark Trust structures that collapse violently when shocks arrive.

Prevention: Pluralism, diversity, and transparency as non-negotiable design constraints.

Summary

Should we redesign economies blind to K and game theory? → No. That’s how you get climate, AI, biosafety, and financial cascades entangled.

Should we redesign them informed by K and coordination laws? → Yes, as: - **Constraints** (don’t drive H_3 toward θ) - **Diagnostics** (spot danger early) - **Design tools** (construct better game forms)

Should K become the single optimization target? → No. K should be a North Star and guardrail, not the only god on the altar.

Naming Convention

Context	Name	Purpose
Academic/Policy	Coordination Political Economy (CPE)	Umbrella field
Formal Framework	K-Stability Economics	Specific toolset
Public/Visionary	Harmonic Political Economy	Books, talks

Example sentence for papers: > “We propose a new coordination political economy in which trust and redundancy are treated as primary factors of production. Within this, we develop a K-Stability framework that uses the K-Index and the Coordination Laws to constrain acceptable policy and technological trajectories.”

Connection to Paper Series

- **Paper 1:** Constructs K-Index empirically
 - **Paper 2:** Applies to historical collapse prediction
 - **Paper 2B:** Modern applications and digital age dynamics
 - **Paper 2C:** Theoretical extensions (Laws 13-17) and cosmic implications
 - **Paper 3 (future):** Full CPE/K-Stability framework for economic design
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Citation

Coordination Political Economy Framework. (2025). K-Stability Economics: Treating Coordination Capacity as Primary Economic Input. [Working document].

Status: Framework document **Last Updated:** December 2025 **Purpose:** Define the economic framework built on coordination physics